

DUNMAN HIGH SCHOOL
2023 H2 PHYSICS (YEAR 6)

2	(a)	Loss in GPE = mgh $= 70.0(9.81)(1.50) = 1030 \text{ J}$	A1
	(b)	kinetic energy of man = $\frac{1}{2}mv^2$ $= \frac{1}{2}(70.0)(2.00)^2 = 140 \text{ J}$	A1
	(c)(i)	The perpendicular distance between the line of action of the weight and the pivot is zero.	B1
	(c)(ii)	The man will fall a further distance equal to the extension of the spring, x before coming to a stop. Assuming no loss in energy to the surroundings, By conservation of energy, loss in k.e. + loss in g.p.e. = gain in e.p.e $(1030 + 140) + 70x = \frac{1}{2} kx^2$ $(1030 + 140) + 70x = \frac{1}{2}(10000)x^2$ Solving: $x = 0.556 \text{ m}$ $\sin \theta = \frac{x}{2.40}$ $\theta = \sin^{-1}\left(\frac{0.557}{2.40}\right)$ $= 13.4^\circ \text{ (to 3 s.f.)}$	M1 M1 M1 A0
		<i>Marker's comments: The board is NOT in rotational equilibrium, i.e. sum of clockwise moments is NOT equal to sum of anti-clockwise moments.</i>	
	(c)(iii)	$F = kx$ $= (10000) (2.40 \sin 13.4^\circ)$ $= 5560 \text{ N}$	A1
	(c)(iv)	The board is in translational equilibrium so net force acting on the diving board is zero. $N = W_{\text{board}} + W_{\text{man}} + W_{\text{spring}}$ $= 300 + 70 (9.81) + 5560$ $= 6550 \text{ N (3 s.f.)}$	C1 A1
		<i>Marker's comments: Students should attempt to provide explanation for their working and not simply show the arithmetic to arrive at the answer.</i>	